

An Introduction to One of the Most Beautiful and Useful Structures in Mathematics

The Symmetric Group is a beautiful algebraic structure that has so many applications across math and the sciences



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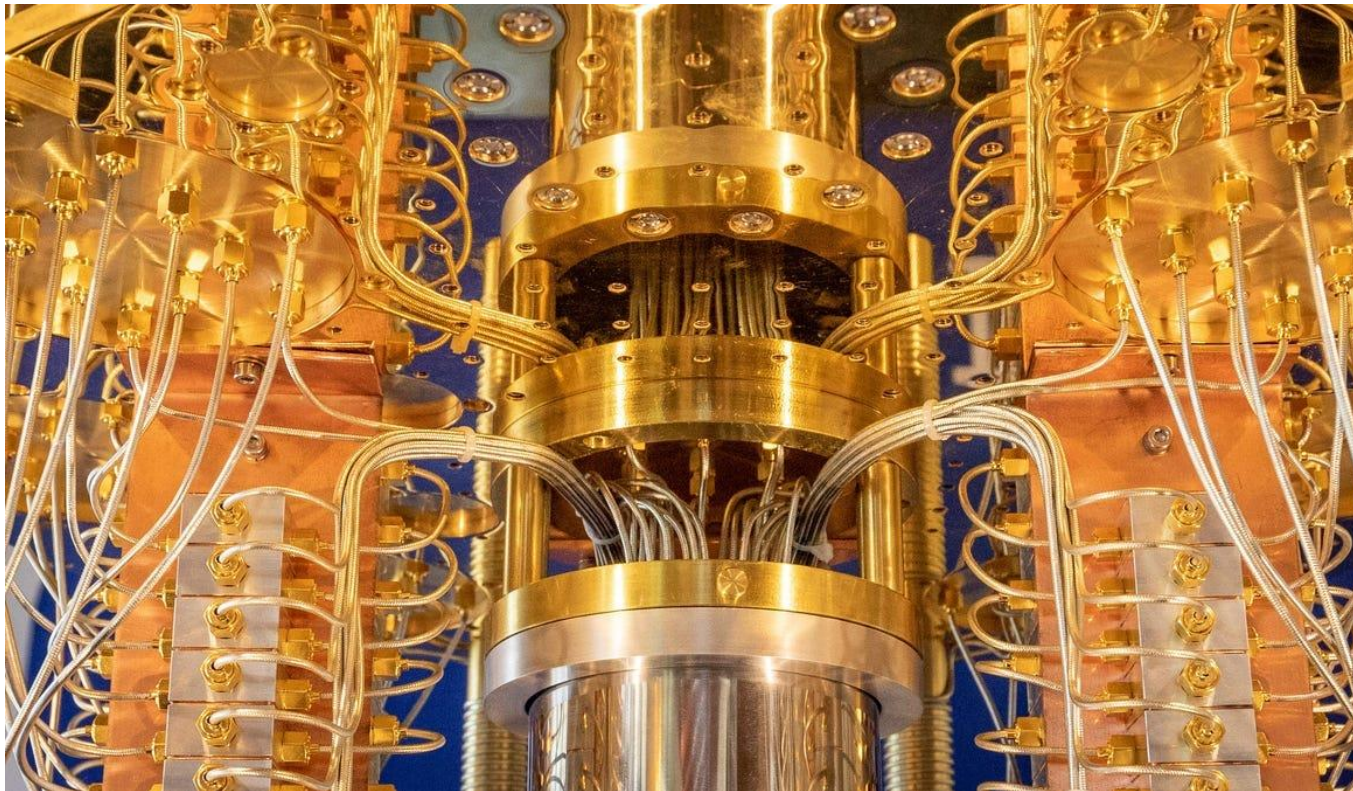
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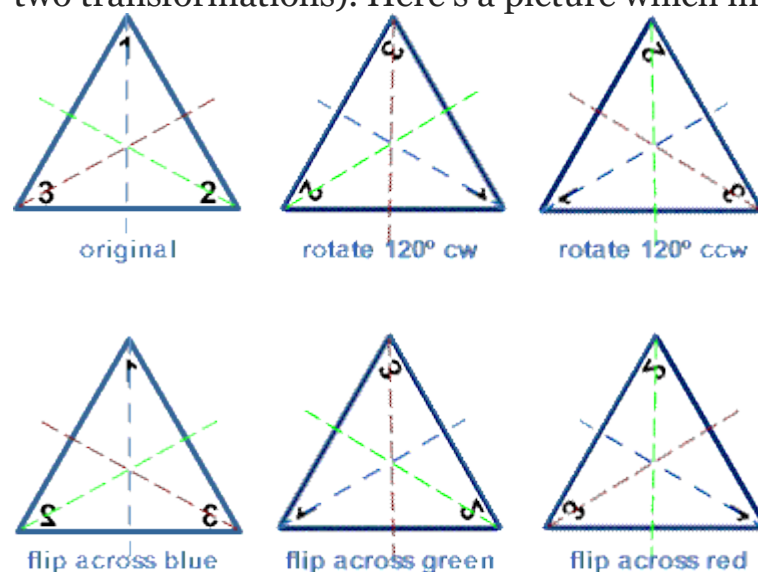
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Quantum computing uses mathematical techniques derived from the Symmetric Group — I discuss this later in the article

What are the ways in which we can transform an equilateral triangle such that it does not lose its shape or orientation? What I mean is: if you cut an equilateral triangle out of a piece of paper, what ways can you move it around such that it will still fit back perfectly in the hole left behind?

A little playing around will tell you that there are six such transformations. Obviously you can use a transformation which does nothing at all (we call this the *identity*), you can flip any two vertices (three transformations, equivalent to reflections in lines of symmetry) and you can rotate the vertices clockwise or anti-clockwise (another two transformations). Here's a picture which might help:



Now, instead of thinking about this geometrically, let's look at it another way. Take the numbers 1, 2, 3 which label the vertices of the first triangle, and let's look at the permutations of these numbers under each of the transformations.

- In the bottom row above, we see that for each flip/reflection, one number stays fixed and the other two swap. We can notate these three flips as (23) , (13) and (12) respectively. This notation tells us where each vertex moves to. So for example, in (23) , 2 moves to where 3 was, 3 moves to where 2 was, and 1 stays fixed.
- The first rotation in the top row moves each vertex one position clockwise. This can be notated (123) — 1 goes to where 2 was, 2 goes to where 3 was and 3 goes to where 1 was. Similarly, the second rotation can be written as (132) , with each vertex moving two positions clockwise.
- If we denote the identity permutation as e , then we have six permutations: e , (12) , (23) , (13) , (123) , (132) . These are all possible permutations of the numbers 1, 2, 3.

The Symmetric Group on n objects

Thought about in this way, we see that the set of symmetries of an equilateral triangle is algebraically equivalent to the set of permutations on three distinct objects. For convenience of notation we usually denote these objects as 1, 2, 3. We call this set S_3 .

Generalizing this, we can denote the set of all permutations of n objects as S_n . We can define an operation on this set by right to left combination. What we mean by this is that for two permutations ρ and σ , the combined permutation $\rho\sigma$ acts by first permuting according to σ and then according to ρ . As an example, if we take (12) and (23) , then $(12)(23)$ acts by first swapping 2 and 3 and then swapping 1 and 2. Careful examination will show that $(12)(23) = (123)$. *[Note: we could also define a left to right combination instead of a right to left combination — that would be mathematically equivalent. If we use this we would have $(12)(23) = (132)$. But right to left is more commonly used by professional mathematicians].*

Algebraically, the set S_n under the combination operation we just described is a *group* in that it obeys the four rules that define a group:

- **Closure:** Any combination of permutations in S_n produces another permutation in S_n
- **Associativity:** For three permutations ρ , σ and τ , we have $(\rho\sigma)\tau = \rho(\sigma\tau)$. For example $(12)(13)(23) = (132)(23) = (12)(132) = (13)$.
- **Existence of identity:** e is the identity permutation
- **Existence of unique inverse:** It should be fairly obvious that for any given permutation, there exists another that restores all objects to their original position. For example $(123)(132) = e$, so (132) is the inverse of (123) and vice versa.

Note that, for $n \geq 3$, S_n is **not** a commutative group. That is, the order in which you combine permutations does matter. For example $(12)(23) = (123)$ but $(23)(12) = (132)$. This is why we had to carefully define the right to left combination action above.

Geometric applications of the Symmetric group

We can see from what we have already done that the entire group S_3 is structurally equivalent to the group of symmetries of an equilateral triangle. In mathematics we say that these two groups are *isomorphic*. Isomorphic groups are basically identical, but just described differently.

We may be easily fooled into thinking, therefore, that S_4 is isomorphic to the set of symmetries of an equilateral rectangle — more usually called a square. But wait, close examination will reveal that a square has 8 symmetries but the number of permutations of 4 objects is $4! = 24$. So these groups cannot be isomorphic. In fact, the group of

symmetries of a square is isomorphic to a subgroup of S_4 . S_4 itself is isomorphic to the symmetry group of a regular tetrahedron.

As we move up the symmetric groups, they get quite large quite quickly, for example S_{10} has $10! = 3,628,800$ elements. These full groups tend not to be useful for describing the symmetries of objects of interest, but subgroups within them are extremely useful — for example, within the symmetric groups you will always find the symmetries of the platonic solids.

This leads us to one of the most powerful facts about the Symmetric Group. *Cayley's Theorem* states that any finite group is isomorphic to a subgroup of S_n . In layman's terms, this tells us that thinking about things in terms of permutations of objects the way we have described above gives us a technology to understand all of finite group theory — a massive and highly applicable field of mathematics. That's quite an amazing thing!

Applications in linear algebra and further afield

One fact about the symmetric group that is extremely useful is that any permutation can be decomposed into a combination of 'swaps'. For example $(123) = (23)(13)$ or $(1234) = (12)(23)(34)$. We can thus define 'odd' and 'even' permutations as follows:

- An odd permutation is a permutation where the shortest decomposition contains an odd number of swaps (so (1234) is an odd permutation)
- An even permutation is a permutation where the shortest decomposition contains an even number of swaps (so (123) is an even permutation)

Now let's look at the determinant of a 2×2 matrix, as follows:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

We learn in high school that

$$\det(A) = a_{11}a_{22} - a_{12}a_{21}$$

Look carefully at those terms. Do you see how the first term (positive) corresponds to the identity element and the second (negative) term corresponds to the permutation (12)? Note also that the identity element is an even permutation and (12) is an odd permutation. So if we define a function on the permutations of S_2 which we call the *sign* function, or $\text{sgn}()$, such that $\text{sgn}(\sigma) = 1$ if σ is even and -1 if σ is odd, we can now express our determinant as:

$$\det(A) = \sum_{\sigma \in S_2} \text{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)}$$

Now let's look at a 3x3 matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

You may have learned similarly in high school that the determinant of such a 3x3 matrix can be calculated by taking a certain linear combination of ‘2x2 subdeterminants’ as follows:

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{23}a_{31})$$

If we expand this, we see that we get six terms corresponding to the six permutations in S_3 , and that the signs correspond to whether the permutations are odd or even. Therefore, as before,

$$\det(A) = \sum_{\sigma \in S_3} \text{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} a_{3\sigma(3)}$$

So you are probably seeing where I am going now. In general, for an $n \times n$ matrix A the determinant of A can be expressed as:

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$$

This is a powerful abstraction allowing the straightforward calculation of determinants of larger matrices. It also abstracts to a broader concept of a *matrix function*, where, for any function f acting on S_n , and for any $n \times n$ matrix A :

$$f(A) = \sum_{\sigma \in S_n} f(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$$

When we use the identity function $f(\sigma) = 1$ for all permutations σ , this matrix function is known as a *permanant*. It represents the direct sum of all permutations of the entries of the matrix. In particle

physics for example, permanents are very important in treating boson Green's functions in quantum field theory, in determining state probabilities of boson sampling systems and in other important topics in quantum computing.

This is a little more advanced for those of you who are familiar, but if the function $f(\sigma)$ is an irreducible character of S_n , these matrix functions are called *immanants*. Because the identity and sign functions are both irreducible characters of S_n , we have that the permanent and the determinant of a square matrix are special cases of immanants. Immanants play important roles in combinatorial chemistry and the study of linear optical networks.

Did you enjoy this brief introduction to the amazing applications of the Symmetric Group S_n ? Feel free to comment!

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